

Shyamoli Textile Engineering College
Department of Computer Science and Engineering
CSE-4237: Digital Image Processing

Lec-16: Wavelet and Multiresolution Processing

Digital Image Processing by Rafael C. Gonzalez and Richard Eugene Woods (Chapter-7)

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3.5 Subband Coding in Wavelet Transform

Subband coding is a method used in signal and image compression where a signal (or image) is decomposed into different frequency subbands.

It is based on multi-resolution analysis using wavelet transforms.

- ❑ It involves splitting a signal/image into multiple frequency bands (subbands).
- ❑ Each subband contains specific low-frequency (coarse details) and high-frequency (fine details) components.
- ❑ The key idea is storing only the important coefficients, which leads to efficient compression.

The Discrete Wavelet Transform (*DWT*) is used for subband coding, where an image is filtered using low-pass and high-pass filters.

Step 1: Decompose the Image

- ✓ Apply a low-pass filter (*L*) to capture low-frequency components (approximate details).
- ✓ Apply a high-pass filter (*H*) to capture high-frequency components (sharp edges and fine details).
- ✓ Downsample (reduce size) to remove redundancy.

Step 2: Generate Subbands

Each level of decomposition results in four subbands:

LL (Low-Low) → Coarse details (low-frequency information)

LH (Low-High) → Horizontal edges

HL (High-Low) → Vertical edges

HH (High-High) → Diagonal edges

Mathematically, for an image $I(x,y)$:

Low-pass (approximation):

$$LL = \sum_x \sum_y I(x, y) \cdot h[2m - x] \cdot h[2n - y]$$

Horizontal detail:

$$LH = \sum_x \sum_y I(x, y) \cdot h[2m - x] \cdot g[2n - y]$$

Vertical detail:

$$HL = \sum_x \sum_y I(x, y) \cdot g[2m - x] \cdot h[2n - y]$$

Diagonal detail:

$$HH = \sum_x \sum_y I(x, y) \cdot g[2m - x] \cdot g[2n - y]$$

where (m,n) are the indices of the transformed coefficients.

Problem:

Compute the 1D DWT of the following signal using the **Haar wavelet**:

$$x=[4,2,5,1]$$

Step 1: Understand the Haar Wavelet Filters

The Haar wavelet uses two filters:

Low-pass filter (scaling function): $h = \left[\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right]$

High-pass filter (wavelet function): $g = \left[\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right]$

For simplicity, we'll use unnormalized filters:

$$h=[1,1], g=[1,-1]$$

Step 2: Convolve and Downsample

The DWT involves convolving the signal with the filters and then downsampling by a factor of 2.

□ Compute Approximation Coefficients (Low-Pass):

$$a[k] = \sum_n x[n] \cdot h[2k - n]$$

For $k = 0$:

$$a[0] = x[0] \cdot h[0] + x[1] \cdot h[1] = 4 \cdot 1 + 2 \cdot 1 = 6$$

For $k = 1$:

$$a[1] = x[2] \cdot h[0] + x[3] \cdot h[1] = 5 \cdot 1 + 1 \cdot 1 = 6$$

So, the approximation coefficients are:

$$a=[6, 6]$$

❑ Compute Detail Coefficients (High-Pass):

$$d[k] = \sum_n x[n] \cdot g[2k - n]$$

For $k = 0$:

$$d[0] = x[0] \cdot g[0] + x[1] \cdot g[1] = 4 \cdot 1 + 2 \cdot (-1) = 2$$

For $k = 1$:

$$d[1] = x[2] \cdot g[0] + x[3] \cdot g[1] = 5 \cdot 1 + 1 \cdot (-1) = 4$$

So, the detail coefficients are:

$$d=[2, 4]$$

Step 3: Final DWT Result

The 1D DWT of the signal $x=[4,2,5,1]$ is:

Approximation Coefficients (Low-Pass) : $a = [6, 6]$

Detail Coefficients (High-Pass) : $d = [2, 4]$

Step 4: Interpretation

The **approximation coefficients** $a=[6, 6]$ represent the low-frequency components of the signal (*averages of pairs of samples*).

The **detail coefficients** $d=[2, 4]$ represent the high-frequency components (*differences between pairs of samples*).

Step 5: Verify with Inverse DWT (Optional)

To verify, we can reconstruct the original signal using the inverse DWT. The inverse Haar transform uses the synthesis filters:

$$\tilde{h} = [1, 1], \quad \tilde{g} = [1, -1]$$

Approximation Coefficients (Low-Pass) : $a = [6, 6]$

Detail Coefficients (High-Pass) : $d = [2, 4]$

Upsample and Convolve:

1. Upsample a and d by inserting zeros:

$$a_{\text{up}} = [6, 0, 6, 0], \quad d_{\text{up}} = [2, 0, 4, 0]$$

Convolve with synthesis filters:

$$x_{\text{reconstructed}} = \tilde{h} * a_{\text{up}} + \tilde{g} * d_{\text{up}}$$

This will give back the original signal $x=[4,2,5,1]$.

Problem:

Compute the 2D DWT of the following 2×2 image: $I = \begin{bmatrix} 4 & 2 \\ 5 & 1 \end{bmatrix}$

Step 1: Understand the Haar Wavelet Filters

The Haar wavelet uses two filters:

Low-pass filter (scaling function): $h = \left[\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right]$

High-pass filter (wavelet function): $g = \left[\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right]$

For simplicity, we'll use unnormalized filters:

$$h = [1, 1], g = [1, -1]$$

Step 2: Apply 1D DWT to Rows

First, we apply the 1D DWT to each row of the image.

$$I = \begin{bmatrix} 4 & 2 \\ 5 & 1 \end{bmatrix}$$

Row 1: [4, 2]

$$h = [1, 1], g = [1, -1]$$

Low-pass (approximation):

$$a_1 = 4 \cdot 1 + 2 \cdot 1 = 6$$

High-pass (detail):

$$d_1 = 4 \cdot 1 + 2 \cdot (-1) = 2$$

Row 2: [5, 1]

$$h = [1, 1], g = [1, -1]$$

$$I = \begin{bmatrix} 4 & 2 \\ 5 & 1 \end{bmatrix}$$

Low-pass (approximation):

$$a_2 = 5 \cdot 1 + 1 \cdot 1 = 6$$

High-pass (detail):

$$d_2 = 5 \cdot 1 + 1 \cdot (-1) = 4$$

After processing the rows, we get:

$$\begin{bmatrix} a_1 & d_1 \\ a_2 & d_2 \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 6 & 4 \end{bmatrix}$$

Step 3: Apply 1D DWT to Columns

Next, we apply the 1D DWT to each column of the row-transformed matrix.

Column 1: [6, 6]

$$h = [1, 1], g = [1, -1]$$

$$\begin{bmatrix} a_1 & d_1 \\ a_2 & d_2 \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 6 & 4 \end{bmatrix}$$

Low-pass (approximation):

$$LL = 6 \cdot 1 + 6 \cdot 1 = 12$$

High-pass (detail):

$$HL = 6 \cdot 1 + 6 \cdot (-1) = 0$$

Column 2: [2, 4]

$$h = [1, 1], g = [1, -1]$$

$$\begin{bmatrix} a_1 & d_1 \\ a_2 & d_2 \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 6 & 4 \end{bmatrix}$$

Low-pass (approximation):

$$LH = 2 \cdot 1 + 4 \cdot 1 = 6$$

High-pass (detail):

$$HH = 2 \cdot 1 + 4 \cdot (-1) = -2$$

Step 4: Final 2D DWT Result

The 2D DWT decomposes the image into four subbands:

$$\begin{bmatrix} LL & LH \\ HL & HH \end{bmatrix} = \begin{bmatrix} 12 & 6 \\ 0 & -2 \end{bmatrix}$$

LL (Low-Low): Approximation coefficients (low-frequency components).

LH (Low-High): Horizontal detail coefficients.

HL (High-Low): Vertical detail coefficients.

HH (High-High): Diagonal detail coefficients.

Step 5: Interpretation

LL = 12: Represents the average intensity of the entire image.

LH = 6: Represents horizontal details (differences between columns).

HL = 0: Represents vertical details (differences between rows).

HH = -2: Represents diagonal details (corner differences).

Problem:

Reconstruct the original image from the 2D DWT result:

$$\begin{bmatrix} LL & LH \\ HL & HH \end{bmatrix} = \begin{bmatrix} 12 & 6 \\ 0 & -2 \end{bmatrix}$$

Step 1: Inverse 1D DWT on Columns

First, we apply the inverse 1D DWT to the columns of the transformed matrix.

Synthesis Filters:

The inverse Haar wavelet uses the following synthesis filters:

$$\tilde{h} = [1, 1], \quad \tilde{g} = [1, -1]$$

Column 1: [LL,HL]=[12,0]

$$\tilde{h} = [1, 1], \quad \tilde{g} = [1, -1]$$

$$\begin{bmatrix} LL & LH \\ HL & HH \end{bmatrix} = \begin{bmatrix} 12 & 6 \\ 0 & -2 \end{bmatrix}$$

Upsample: Insert zeros between the coefficients:

$$[12, 0] \rightarrow [12, 0, 0, 0]$$

Convolve with $\tilde{h}$:

$$\text{Row 1} = 12 \cdot 1 + 0 \cdot 1 = 12$$

$$\text{Row 2} = 12 \cdot 1 + 0 \cdot (-1) = 12$$

Column 2: [LH,HH]=[6,-2]

$$\tilde{h} = [1, 1], \quad \tilde{g} = [1, -1]$$

$$\begin{bmatrix} LL & LH \\ HL & HH \end{bmatrix} = \begin{bmatrix} 12 & 6 \\ 0 & -2 \end{bmatrix}$$

Upsample: Insert zeros between the coefficients:

$$[6, -2] \rightarrow [6, 0, -2, 0]$$

Convolve with $\tilde{h}$:

$$\text{Row 1} = 6 \cdot 1 + (-2) \cdot 1 = 4$$

$$\text{Row 2} = 6 \cdot 1 + (-2) \cdot (-1) = 8$$

After processing the columns, we get:

$$\begin{bmatrix} 12 & 4 \\ 12 & 8 \end{bmatrix}$$

Step 2: Inverse 1D DWT on Rows

Next, we apply the inverse 1D DWT to the rows of the column-transformed matrix.

Row 1: [12,4]

$$\tilde{h} = [1, 1], \quad \tilde{g} = [1, -1]$$

$$\begin{bmatrix} 12 & 4 \\ 12 & 8 \end{bmatrix}$$

Upsample: Insert zeros between the coefficients:

$$[12, 4] \rightarrow [12, 0, 4, 0]$$

Convolve with $\tilde{h}$:

$$\text{Column 1} = 12 \cdot 1 + 4 \cdot 1 = 16$$

$$\text{Column 2} = 12 \cdot 1 + 4 \cdot (-1) = 8$$

Row 2: [12,8]

$$\tilde{h} = [1, 1], \quad \tilde{g} = [1, -1]$$

$$\begin{bmatrix} 12 & 4 \\ 12 & 8 \end{bmatrix}$$

Upsample: Insert zeros between the coefficients:

$$[12, 8] \rightarrow [12, 0, 8, 0]$$

Convolve with $\tilde{h}$:

$$\text{Column 1} = 12 \cdot 1 + 8 \cdot 1 = 20$$

$$\text{Column 2} = 12 \cdot 1 + 8 \cdot (-1) = 4$$

Step 3: Normalize the Result

The inverse DWT results in scaled values. To recover the original image, we divide by the scaling factor. For the Haar wavelet, the scaling factor is $1/2$.

$$\text{Reconstructed Image} = \frac{1}{2} \cdot \begin{bmatrix} 16 & 8 \\ 20 & 4 \end{bmatrix} = \begin{bmatrix} 8 & 4 \\ 10 & 2 \end{bmatrix}$$

Step 4: Compare with Original Image

The original image was: $I = \begin{bmatrix} 4 & 2 \\ 5 & 1 \end{bmatrix}$

The reconstructed image is: $\begin{bmatrix} 8 & 4 \\ 10 & 2 \end{bmatrix}$

Step 5: Fix the Scaling Issue

The discrepancy arises because we used unnormalized filters. To fix this, we normalize the filters by dividing by $\sqrt{2}$:

$$h = \left[\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right], \quad g = \left[\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right]$$

Repeating the calculations with normalized filters will yield the exact original image:

$$I = \begin{bmatrix} 4 & 2 \\ 5 & 1 \end{bmatrix}$$